

Identifying modern slavery in global supply chains: Leveraging monitoring technologies through multi-actor collaboration

Cory Searcy^{a,*}, Pavel Castka^b, Grant Michelson^c, Xiaoli Zhao^b

^a Toronto Metropolitan University, Toronto, Canada

^b Xi'an Jiaotong - Liverpool University, Suzhou, China

^c Macquarie University, Sydney, Australia

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ABSTRACT

Monitoring technologies (e.g., worker voice technologies, remote sensors, satellite images) provide additional opportunities to improve the identification of modern slavery in supply chains. New collaborations among different actors are required to enable these technological capabilities. Yet little is understood about how collaboration between actors such as employers, certification and auditing bodies, non-governmental organizations (NGOs), and other vested intermediaries leverage monitoring technologies to identify modern slavery in global supply chains. Based on a qualitative inquiry of 32 interviews with leading actors in identifying modern slavery, we build upon two domains in the literature: the contracts domain and the conditions domain. Drawing on resource dependence theory (RDT), we show that valued resources (finance, access, skills, technology, and legitimacy) are held by various interdependent collaborators. We further show that identifying modern slavery can be enhanced through leveraging monitoring technologies embedded in collaborations, yet key contingencies (modern slavery posture, collaboration scope, cross-boundary interactions, and contextual embeddedness) facilitate (or inhibit) such an outcome. The paper offers an empirically grounded understanding of collaboration mechanisms to detect modern slavery in global supply chains.

1. Introduction

Many companies have increased their efforts to identify modern slavery, supported by the growth of modern slavery and due diligence legislation around the world. However, the illegal nature of modern slavery, the difficulty of recognizing it in practice, and debates about the effectiveness of existing initiatives suggest legislation alone is insufficient (Stevenson and Cole, 2018). Modern slavery is a contested term, but it fundamentally concerns the “severe exploitation of other people for personal or commercial gain”.¹ Severe exploitation remains all too common, with millions of people facing controls and limits on their freedom, harsh working conditions, and physical and emotional abuse. The International Labour Organization (ILO) estimates that as many as 50 million people remain in modern slavery around the world (ILO, 2022, p. 2).

Efforts to eliminate modern slavery in supply chains are increasingly being met by multi-actor collaborations (Benstead et al., 2018, 2021). While there can be differences between supply chains, these can involve collaborations between employers, certification and auditing bodies, non-governmental organizations (NGOs), and other vested intermediaries. However, the results are often poor, and even organizations that explicitly focus on addressing modern slavery find it challenging to tackle. For example, the chocolate company Tony’s Chocolonely has long been committed to eliminating child labour, forced labour, and human trafficking in its cocoa supply chain, yet it acknowledges the sector remains “full of human rights violations”.² The experience of Tony’s Chocolonely underlines the need for collaboration where all key actors take responsibility for eliminating abuses in the supply chain (Pannekoek et al., 2023). Innovative approaches recognizing the need for enhanced collaboration are urgently required to

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* Corresponding author.

E-mail addresses: cory.searcy@torontomu.ca (C. Searcy), Pavel.Castka@xjtlu.edu.cn (P. Castka), grant.michelson@mq.edu.au (G. Michelson), Lilly.Zhao@xjtlu.edu.cn (X. Zhao).

¹ <https://www.antislavery.org/slavery-today/modern-slavery/>.

² <https://us.tonyschocolonely.com/pages/our-promise#sectionid-2>.

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identify modern slavery in global supply chains.

There is a growing body of research demonstrating that technologies can help identify modern slavery in supply chains. Technologies discussed in a modern slavery context typically focus on those that enable monitoring of organizational practices and/or traceability of products rather than technologies assisting in direct operational processes. Examples include technologies enabling traceability (McGrath et al., 2021), blockchain technologies (Christ and Helliard, 2021), worker voice technologies (Berg et al., 2020; Taylor and Shih, 2019), or satellite imaging technologies (Koukoulou et al., 2021). We label these technologies as “monitoring” technologies which can improve data collection, data exchange, and data analytics to improve the identification of modern slavery (Cossio and McNeill, 2021). This paper explores how multi-actor collaborations leverage monitoring technologies to enhance the likelihood of identifying modern slavery. Solutions grounded in collaborative approaches to modern slavery offer considerable promise (Benstead et al., 2021), but little is currently understood about when, why, and how these collaborations occur and, critically, how monitoring technologies are leveraged through multi-actor collaborations to identify modern slavery.

Our study is grounded in two perspectives. First, we build on previous literature in modern slavery with a focus on two domains within it: the *contracts-focused domain* and the *conditions-focused domain* (Simpson et al., 2021). Each domain exhibits unique characteristics that impact the choice of both collaborators and monitoring technologies. Second, our study utilizes resource dependence theory (RDT). RDT is a highly influential theory (Hillman et al., 2009; Ozturk, 2021) that seeks to understand control between firms from two perspectives: power imbalance and mutual dependence (Shiu et al., 2023). In this study, we draw on the mutual dependence perspective. The mutual dependence perspective can be used to explain when and why different organizations collaborate with each other to identify modern slavery. Such motivations might include strengthening organizational power, autonomy and societal acceptance, or legitimacy (Drees and Heugens, 2013). Importantly, the mutual dependence perspective recognizes that important and necessary resources are held by different collaborators and that there are interdependencies between those collaborators and the resources they control (Pfeffer and Salancik, 1978). We do not focus on the power imbalance perspective. While it may be argued that mutual dependence and power imbalance are connected, our primary interest is explaining collaboration rather than competition, as may be implied through attention given to power imbalance. While RDT is typically concerned with explaining firm survival rather than addressing or eliminating injustice, it also provides a lens on the critical role resources play in addressing potential reputational and financial risk. Such risks can harm firm well-being and, indirectly, their long-term survival, if injustices, like modern slavery, are found. With research showing that firms can financially benefit if they have a track record of addressing modern slavery risk (Cousins et al.), identifying modern slavery through collaborative means can be an important step towards risk mitigation and firm success.

Building on the above, our research question is: *How can multi-actor collaborations leverage monitoring technologies to identify modern slavery in global supply chains?* While there are a range of actors that are potentially relevant in the context of modern slavery including unions, consumers and investors (McLaren et al., 2024), we focus particularly on collaborations between employers, certification and auditing bodies, NGOs, and other vested intermediaries. Consistent with RDT, we assume that organizations pursue collaborations to address gaps in their capabilities and/or capacity. This underscores that organizations require resources (e.g., data, access, legitimacy, and technology) to facilitate their operation, reduce uncertainties and risk, and ultimately ensure their survival. Since few organizations are self-sufficient and leveraging monitoring technologies to identify modern slavery can require unique competencies, they must collaborate with other organizations to meet capability gaps.

To address our research question, we adopt a qualitative approach. We conducted 32 interviews with different organizations involved in addressing modern slavery, inclusive of NGOs, business brands, auditing firms, individual auditors, technology providers, and regulatory bodies. All respondents had extensive experience in the domain of modern slavery (such as lawyers, managers, activists, purchasing managers, governmental officials), in different regions of the world. The respondents had considerable experience in multi-actor collaboration and in the use of technologies in the context of modern slavery and covered a wide range of industries. Building on previous studies on modern slavery and drawing on RDT to frame and interpret our findings, we demonstrate that leveraging monitoring technologies in multi-actor collaboration to identify modern slavery is facilitated (or inhibited) by four key contingencies: modern slavery posture, collaboration scope, cross-boundary interactions, and contextual embeddedness.

2. Conceptual and theoretical background

2.1. Modern slavery and supply chains

Modern slavery is often manifest through indicators such as forced work under threat or harm to the worker, owner/employer control (e.g., retention of identity documents or passports), restrictions on freedom of movement, financial mistreatment (e.g., withholding of wages, underpayment), and a general dehumanization and commoditization (Caruana et al., 2020; Crane, 2013, p. 53). While different economic and institutional conditions can precipitate modern slavery (Moussa et al., 2022), there are important capabilities that firms might utilize to insulate, sustain, and profit from such conditions (Crane, 2013). Indeed, it has been argued there are multiple business models that can perpetuate modern slavery, including models grounded in risk reduction, asset leveraging, evading legal minimums, and workers as consumers (Crane et al., 2022). Responsibility for the presence of modern slavery rests, at least partly, with firm managers and it is within complex organizational supply chains where such exploitative practices might be most apparent and yet most hidden (Gold et al., 2015; LeBaron, 2021; New, 2015; Shilling et al., 2021; Stringer and Michailova, 2018).

Legislative developments aimed at eliminating modern slavery across numerous jurisdictions along with NGO vigilance have increased scrutiny and reputational risk in global supply chains (Nolan and Bott, 2018; Raj-Reichert, 2020). Among the growing examples are the recent Uyghur Forced Labour Prevention Act in the United States and the Supply Chain Management Act as part of due diligence legislation in Germany. While there is debate about the effectiveness of some of these laws, they do raise modern slavery's public profile (Islam and Van Staden, 2022). Moreover, discovery of modern slavery can heighten investor concerns around risk and negatively impact the market value of the affected organization (Nunes, 2018) suggesting that greater transparency in supply chains is beneficial (Kraft and Zheng, 2021). Despite these advantages, research has indicated that modern slavery is a “psychologically distant” or remote idea for many firms (Simpson et al., 2021).

Two domains have been identified to help structure efforts to reduce “psychological distance” from modern slavery (Simpson et al., 2021): the *contracts-focused domain* and the *conditions-focused domain*. The *contracts-focused domain* focuses predominantly on confirming the absence of modern slavery (Ibid). This domain is audit centric, with an emphasis on comparing performance relative to a reference point. In contrast, the *conditions-focused domain* is driven by the context in which modern slavery occurs and focuses predominantly on identifying the presence of modern slavery (Ibid). This domain is typically investigative in nature, with the mindset being to find, disclose, and remediate modern slavery in supply chains. However, even when modern slavery is found, firms might act defensively regarding the uneven or ambiguous public disclosure of the effectiveness of their actions against modern slavery (Meehan and Pinnington, 2021; Schaper and Pollach, 2021;

Stevenson and Cole, 2018), or there can be inaction and the failure to acknowledge its occurrence (Stevenson, 2022).

The contracts- and conditions-focused domains provide a granular understanding of how to identify modern slavery, recognizing how different approaches can lead to different outcomes. However, much like firms can simultaneously adopt symbolic and substantive practices to address non-financial issues such as quality or sustainability issues (Prajogo et al., 2021), they can also simultaneously adopt approaches that fall within and bridge the contracts- or conditions-focused domains. Little is currently understood, however, about how the two domains co-exist and how activities within each domain interact. This is important given the need to understand how different collaborations can leverage different monitoring technologies within and across the two domains to identify modern slavery.

2.2. Collaboration to identify modern slavery

Collaboration occurs when “two or more independent companies work jointly to plan and execute supply chain operations with greater success than when acting in isolation” (Simatupang and Sridharan, 2002). Given such a broad definition, it is clear that multiple forms of collaborations exist in global supply chains, from strategic alliances to more arms-length collaborations (Hayes et al., 2005). Moreover, collaborations are driven by context: there is no single best way or blueprint on how different actors collaborate. Fundamentally, collaborations are pursued to increase success of involved actors in achieving their objectives. These could include, for example, instrumental objectives focused on identifying modern slavery to prevent firm reputational damage. Collaboration can also be altruistic in nature, recognizing the moral obligation to remedy severe worker exploitation (Pannekoek et al., 2023).

The main types of supply chain collaboration strategies are horizontal and vertical integration (Barratt, 2004). Horizontal collaboration occurs when two or more competing or unrelated firms (at the same level of the supply chain) cooperate to share resources, such as manufacturing capacity. Horizontal collaboration has been examined in response to modern slavery and requires relational capital and effective governance mechanisms to succeed (Benstead et al., 2018). Vertical collaboration occurs at different levels of the supply chain – with customers, internally (across functions), service providers, and with suppliers (Barratt, 2004). Collaborations between different producers, distributors and other actors enables a more contextualized approach in supply chains (Caputo and Mininno, 1996), arguably providing more scope to strengthen labour standards and conditions (LeBaron, 2021).

Collaboration is an area of increasing focus in sustainable supply chain research including the need for cross disciplinary scholarship (Kach et al., 2025), but social concerns, such as extreme labour conditions, have rarely been studied in this context (Geng et al., 2017). Much of the limited research that exists focuses on relationships between buyers and suppliers, suppliers for a common buyer (Touboulie and Walker, 2015), or buyers with common suppliers (Nidumolu et al., 2014). Collaborative practices to identify modern slavery are critical, as they do not just impact source firms but can enhance the capabilities of other actors to reduce modern slavery risks, such as suppliers (Emberson et al., 2022). Suppliers may adapt their vertical and horizontal relationships to reduce the risk of modern slavery through strategies such as horizontal systemization, vertical systemization, horizontal transformation, and vertical differentiation (Emberson et al., 2022). The implication is that collaboration recognizes and promotes agency among the different actors to enact a range of complementary and mutually beneficial outcomes for all.

Collaborations beyond supplier and buyer relationships, particularly in a modern slavery context, can involve direct interventions, awareness campaigns, or government advocacy for policy changes, among others. Collaboration to combat modern slavery typically starts by building coalitions where firms, NGOs, and government agencies mobilize

around a common goal (Gutierrez-Huerter et al., 2023). Many initiatives focus on raising awareness to help prevent exploitation (Bales, 2012), or on training and capacity building. Corporations, NGOs, and governments often work together to train law enforcement, immigration officials, and social service providers on human trafficking issues (Raineri and Strazzari, 2022). Collaboration on supplier audits is another form of collaboration. Based on a study of 35 brands, an NGO, and a trade body, Benstead et al. (2018) investigated how horizontal collaboration can enable responses to modern slavery legislation in the textiles and fashion industry. Franssen and LeBaron (2019) examined the role of the “Big Four” audit firms in services connected to transnational labor governance. Drawing on a case study linked to modern slavery, they found that auditing firms “engage in a variety of informal and covert influencing practices”, such as in promoting incremental improvements to soft-law labor governance and serving as civil society’s independent watchdog (Franssen and LeBaron, 2019, p. 260). Gold et al. (2015) also highlighted the need for multi-actor approaches to address modern slavery.

2.3. Using technology to identify modern slavery in supply chains

Technology can be broadly defined as “the practical application of knowledge especially in a particular area” or “a manner of accomplishing a task especially using technical processes, methods, or knowledge”, among many other possible formulations.³ In this paper, we focus specifically on monitoring technologies. Monitoring technologies can be classified as a form of “indirect” technologies. Unlike direct technologies (i.e., technologies that enable transformation of inputs into outputs, such as trucks, flexible manufacturing systems or material processing technologies), indirect technologies focus on controlling and coordinating direct processes. Technology has been used to support broader social compliance audits, with a recent Gartner report providing a set of categories to structure thinking about its potential application (Cossio and McNeill, 2021).

Table 1 groups monitoring technologies into three main categories: technologies enabling data collection, data exchange, and data analytics. First, monitoring technologies can be used for *data collection*, either on-site or in a specific geographic area. For example, landlines and video cameras are frequently used at the facility-level. Data collection can be enabled by the use of satellite imaging in a broader geographic area (Reid and Castka, 2023). Second, monitoring technologies can be used for *data exchange*. Technologies can be used to ensure interoperability between actors, which is important for any form of collaboration (Herding and Fischer, 2015). At a facility, for example, smart phones might be used by auditors to upload data in shareable formats to apps connected to broader audit platforms (Castka and Searcy, 2023). Blockchain can then provide an unalterable ledger to provide confidence on product provenance throughout a supply chain. Third, monitoring technologies can be used to help draw conclusions about the data collected through *data analytics*. For example, algorithms could be developed to predict modern slavery hotspots. Research on the specific use of technologies to identify modern slavery has started to emerge, including the use of technology to improve product traceability (McGrath et al., 2021), using blockchain (Christ and Helliari, 2021), applying worker voice technologies (Berg et al., 2020; Taylor and Shih, 2019), and applying satellite imaging (Kougkoulos et al., 2021). More broadly, institutional pressures have been shown to encourage digital technology adoption by firms seeking to alleviate modern slavery risks (Jiang et al., 2023).

Technology is being increasingly incorporated to facilitate collaborations between different actors and optimize the use and allocation of resources (Castka and Searcy, 2023). These collaborations are usually initiated by a focal actor needing resources to accomplish their

³ <https://www.merriam-webster.com/dictionary/technology>.

Table 1

Illustrative examples of using technology to identify modern slavery in supply chains.

Category	Illustrative examples of monitoring technologies and their use
Data collection	- Supply chain mapping technologies: platforms and tools that help companies visualize their supply chains based on data gathered on supplier locations, relationships, and product flows. - Worker voice technologies and hotlines: facilitate direct reporting from workers on abuse, exploitation, and unsafe working conditions by collecting real-time data. - Auditing technologies: use of digital platforms to manage audit data, including worker interviews, facility inspections, and document reviews. - Satellite monitoring and remote sensing: collection of imagery and sensor-based data related to deforestation, illegal mining, and other activities that may be linked to modern slavery. - Open data and publicly available resources: gathering data from government reports, NGO databases, social media, and news articles to identify risk factors.
Data exchange	- Blockchain technology: facilitates secure and transparent data exchange between different actors in the supply chain; allows for tracking products from origin to consumer and verifying claims related to ethical sourcing. - Supplier ethical data exchange: enables companies to share responsible sourcing data with their suppliers and customers (e.g., Sedex). - Cloud-based platforms: allows companies to share information about suppliers, audit results, and other relevant data with stakeholders, potentially in real-time.
Data analytics	- Big Data and AI-powered analytics: assemble large datasets from various sources (e.g., supply chain mapping, audits, worker reports, news articles, etc.) to analyze data and identify high-risk suppliers or regions based on various indicators. - Natural Language Processing: use of NLP to analyze text data (e.g., news articles, social media posts) to identify potential cases of modern slavery. - Geospatial analysis: analyze geographical data to identify areas with high concentrations of vulnerable populations or industries known for forced/exploited workers.

objective, but there are several potential barriers to collaboration that seek to leverage technology. Different technologies used by suppliers throughout a supply chain can be an important barrier to collaboration, but many potentially “intractable” barriers are rooted in people issues, such as “culture, trust, aversion to change, and willingness to collaborate” (Fawcett et al., 2008). Overcoming these barriers is essential to successfully leverage monitoring technologies in multi-actor collaborations. For example, in a modern slavery context, an auditing firm may collaborate with NGOs for access to genuine employee perspectives or local communities (Taylor and Shih, 2019). Technologies play a role in addressing collaborative challenges that exist between these actors. For instance, actors involved in supplier auditing are exposed to legitimacy risks due to poor timeliness and veracity in supplier auditing (Castka et al., 2020). Data collection can be enabled through the use of mobile apps (Locke, 2013), which allow for faster data exchange. Mobile apps and other technologies also permit the collection of greater amounts of data – in part mitigating bias (Solymosi and Bowers, 2018). Machine learning and artificial intelligence (AI) can be used to identify patterns of trafficking and slavery in large datasets, making it easier to identify at-risk areas or at-risk individuals (Land and Barry, 2021). Technology could thus help NGOs, firms, and regulatory agencies to share pertinent data and work together more effectively.

Technology, however, is not a panacea. There is potentially a dark side to the use of monitoring technology, such as strengthening the ability to control employees through surveillance or serving as a cover for minimal or no performance improvement (Mol, 2010). Research is also emerging on the potential dissatisfaction in adopting supply chain technologies, such as blockchain, the Internet of Things (IoT), and artificial intelligence (AI) (Sodhi et al., 2022). As Sodhi et al. (2022, p. 2517) explain, the perceived benefits “depend on the interaction between technology and the users, not on the technology alone”. This point

is supported by Marques et al. (2024), who demonstrated that, while blockchain is often promoted as a solution to supply chain transparency problems, this can be substantially undermined by a lack of trust between supply chain partners. Blockchain can also introduce additional unanticipated costs and risks (Murray et al., 2021). Technology can thus create “inflated expectations” (Sodhi et al., 2022) in supply chain management, including in the ability to identify modern slavery. Successfully leveraging monitoring technology to identify modern slavery is therefore contingent on multi-actor collaborations grounded in trust, a genuine willingness to collaborate, and a recognition of its limitations.

2.4. Theoretical underpinnings

There are many actors (e.g., businesses, service providers, NGOs, auditors, employees, etc.) as identified by McLaren et al. (2024), and monitoring technologies (e.g., blockchain, remote sensing and satellites, worker voice applications, etc.) that may be necessary to identify modern slavery. While institutional theory and global value chain theory are the most frequently used theories in modern slavery and supply chains research (Strand et al., 2024, p. 6), we pursue a different line of inquiry. We draw on resource dependence theory (RDT) as a lens for our study. RDT understands control between firms from two perspectives: power imbalance and mutual dependence (Shiu et al., 2023). While power imbalances exist within supply chains and can impact the ability to identify modern slavery, we regard this perspective as highly suited to accounting for competitive situations. Our focus in this study is on the mutual dependence perspective and its ability to explain multi-actor collaboration. Previous research has shown that mutual dependence has a positive impact on collaborative performance (Casciaro and Piskorski, 2005; Shiu et al., 2023). Therefore, consistent with Shiu et al. (2023, p. 2), we adopt the view that “a firm needs to devote time and resources towards coordinating other firms in order to obtain resources for its strategic goals”. The mutual dependence perspective underpins our theoretical approach, since it emphasizes the need for multiple actors, often spread across multiple jurisdictions, to work together to identify modern slavery.

RDT is primarily concerned with understanding and explaining firm survival. RDT argues that for an organization to survive, it must acquire valued and necessary resources, but that these resources are often held by others (Pfeffer and Salancik, 1978). This creates uncertainty for organizations, precipitating them to manage their environmental uncertainty and resource acquisition in different ways. Resources can be tangible or intangible assets that are viewed as organizational strengths (Wernerfelt, 1984). Resources could be direct, such as labour and raw materials, or they could be indirect, such as favourable legislation. Resources could also be considered from the perspective of substantive resources, such as cash flow and technology, or symbolic resources, such as legitimacy and reputation (Meyer and Rowan, 1977). Different organizations hold and control different resources and, to mitigate risks and survive, organizations must interact with other entities which control them (Scott and Davis, 2015).

Organizations are conceptualized as open systems embedded in a network of interdependencies with other organizations (Pfeffer and Salancik, 1978). The interdependence with other organizations lies in the availability of different (valued) resources and demand for them. For example, service providers require clients to use the technology they are selling to help identify modern slavery. NGOs require resources to fund their social objectives but also appreciate that firms value the legitimacy they can provide (Michelson and Wray-Bliss, 2024). RDT helps explain which actors (and why) help provide valued resources – including monitoring technologies – at particular points in time (the when). Moreover, collaboration shows how different dependencies in modern slavery are managed. If we consider that modern slavery can threaten firms, specifically by exacerbating reputational and financial risk, the use of monitoring technology can either illuminate or conceal modern slavery practices (Marques et al., 2024). It can therefore subsequently

shape how collaborations occur – with whom, when, and why. The capacity of organizations seeking to identify modern slavery therefore depends on the available resources. We posit that such resources can be acquired and subsequently deployed through collaborative means, and is one important step towards risk mitigation and firm survival.

Some scholars have contended that RDT has confounded the separate dimensions of power imbalance or differentials between organizations and the idea of mutual dependence on each other in the construct of interdependence (Casciaro and Piskorski, 2005). That is, organizations may leverage their extant power to increase their influence on others over time, thereby reducing their dependency (Shen et al., 2022). There is some merit to this claim, but as explained earlier our focus is primarily on mutual dependence. Nonetheless, we note that organizations with more power than others need not exercise or impose this power, especially when seeking legitimacy or societal acceptance around the issue of modern slavery. Organizations can build trust over time, especially when there are asymmetries in power and other resources between actors that collaborate (Gulati and Sytch, 2007; Mosch et al., 2021). A meta-analysis of previous studies that have used RDT confirms this tendency by showing if organizations are primarily interested in improving their legitimacy, they create interorganizational collaborations that create visible linkages with other reputable organizations; if organizations are primarily motivated to improve their autonomy, they create collaborations with other organizations that preserve their flexibility (see Drees and Heugens, 2013). Moreover, as Shiu et al. (2023, p. 9) have explained, “firms should not only manage their mutual dependencies but also consider how their actions may affect indirect interfirm relationships beyond the direct mutual dependence itself”.

It is important to acknowledge that dependence within a RDT framework is sometimes considered undesirable (Ozturk, 2021, p. 547). However, there are certain costs with maintaining independence and autonomy when different organizational actors try to identify modern slavery. We argue that collaborations are fundamentally connected to purposes grounded in RDT. Collaborations ultimately come down to accessing valued resources held by others. In a modern slavery context, these resources can include financing, access, skills, and legitimacy. Each resource has a unique role in the process of identifying modern slavery. For example, skills (to perform auditing and investigation) are critical to identify modern slavery. Actors can (potentially) optimize the use of these skills (i.e., auditing skills, analytics skills, etc.) through collaboration and thus improve the likelihood of identifying modern slavery. Critics might contend that this may perpetuate an instrumental, rather than ethical, approach to the identification of modern slavery. However, the meta-analysis of RDT by Drees and Heugens (2013) confirms that the theory can also account for firm actions that are motivated by meeting societal norms and expectations as opposed to only addressing financial performance (survival). In this context, technologies can further leverage other resources. For example, monitoring technologies can support auditors on the ground (e.g., wearable computers, apps for data collection) or can facilitate access to a larger pool of vulnerable respondents (e.g., through worker voice technologies). A representative overview of resources and their role in identifying modern slavery is shown in Table 2.

The necessity of managing interorganizational relations is particularly salient in global supply chains (Mosch et al., 2021) given the prevalence of power and dependence noted above. Collaboration can bring together the resources that help firms achieve their strategic goals (Shiu et al., 2023), in this case identifying modern slavery. While collaborations and resources can change over time, the ability to identify modern slavery is grounded in mutual dependence. This is particularly salient in situations where formal authority is unlikely to be sufficient to address a highly complex problem. Identifying modern slavery requires substantial social capital, including the need for collective action to build trust, legitimacy, and capacity in difficult environments. The literature shows that mutual dependencies are more likely to emerge when firms cannot rely on formal authority and must build social capital

Table 2

Illustrative examples of resources for identifying modern slavery.

Resource	Importance of the resource for identifying modern slavery	The potential of collaboration to leverage the resource	The potential of technology to leverage the resource
Financing	Resourcing of audits and investigations is critical	Collaborations can optimize allocation of financial resources to operationalize audits and investigations to identify modern slavery	Technologies can provide tools to increase efficiency of audits (i.e., audit platforms utilize data in multiple audits and address the issue of duplication of audits)
Access	Access to local environments is critical for identifying modern slavery	Collaborations can provide access to local environments (e.g., access to local informants) to improve the potential for identifying modern slavery	Technologies facilitate access to respondents in locations (e.g., hotlines) and/or allow scaling up communication with local respondents (e.g., worker voice technologies)
Skills	Auditing and investigative skills are an essential mechanism for identifying modern slavery	Collaborations can optimize the use of unique skills possessed by each actor (e.g., auditing skills, analytics skills, etc.) to improve the potential for the identification of modern slavery	Technologies provide support for actors to utilize their skills (e.g., auditors use wearable computers or use apps for data collection) and/or facilitate access to local data (e.g., data platforms can use analytics to determine hotspots)
Legitimacy	Audits and investigations need to be perceived as legitimate	Collaborations can increase legitimacy of audits and investigations	Technologies can improve veracity and timeliness of audits and, hence, potentially strengthen the legitimacy of audits and investigations

(Shiu et al., 2023).

3. Methods

3.1. Research approach

Studies of modern slavery in supply chains remain in their embryonic stage and often struggle to secure access to robust and representative data. Modern slavery is also difficult to measure directly and it is often manifested through proxy indicators requiring significant judgment.

Our research question focused on how multi-actor collaborations can leverage monitoring technologies to identify modern slavery. We are not focused on any particular monitoring technology (such as worker voice technologies or blockchain which was the objective of Jiang et al., 2023); our aim is to explore how collaboration between actors such as employers, certification and auditing bodies, NGOs, and other vested intermediaries might leverage these technologies to support identifying modern slavery in global supply chains. Our point of departure is that the research is best addressed by those with direct experience in identifying modern slavery. Our approach was guided, therefore, by gathering detailed insights from experts in the field through a qualitative study.

3.2. Data sources and data collection

We approached experts that held leading roles in institutions with

global outreach. For example, we spoke with purchasing directors directly involved in developing approaches to identify modern slavery in their complex supply chains. We also targeted experienced NGO directors who have overseen projects with numerous collaborators, including corporations, their suppliers, and other NGOs. We recruited senior experts responsible for overseeing certifications bodies and their activities. These individuals have access to numerous audit reports, data

on grievances, and non-compliance, which provides them with unique insight into the problem. Overall, we recruited individuals directly involved in identifying modern slavery, particularly through using monitoring technology leveraged through collaborations.

While Table 3 lists the experts' role at the time of the study, many had progressive experience in a variety of roles directly related to our research question. For instance, while there were a large number of

Table 3

List of participants and interview details (n = 32).

Actor Group	Role	Outreach in global supply chains	Years with the organization	# of interviews	Code
NGOs	Co-director	Co-leading a coalition of 30 plus organizations	12 years	2	MS1, MS5
	Co-director	Co-leading a coalition of 30 plus organizations	11 years	2	MS2, MS4
	Program Director	A knowledge platform addressing cross-boundary interactions; working mainly with financial services, apparel & footwear, hospitality providers and retailers	4 years	1	MS6
	Director	A leader in technology adoption to monitor human trafficking	7 years	1	MS11
	Sustainability Director	Provides research, training, consulting, and policy advocacy across a variety of sectors globally in over 70 countries	2 years	1	MS14
	Managing Director	A global NGO focusing on anti-trafficking projects concentrated in technology, financial crimes, and regulatory frameworks	11 years	1	MS21
	Chief Impact Officer	A global NGO operating across three continents in the conditions-domain	6 years	1	MS22
	Chief Technical Advisor	An agency bringing together governments, employers and workers addressing cross-boundary interactions	10 years	1	MS25
	Senior Advisor	Focusing of seafood industry; leads collaborative projects across the Asia Pacific region	3 years	1	MS30
Firms	Technical and Projects advisor	A global actor providing education, advocacy and third-party assurances in the seafood sector	18 years	1	MS32
	Director of Sustainability	A global multinational company from the food industry with an extensive network of partners across both domains	5 years	1	MS10
	Sustainability Director	A global omni-channel retailer overseeing modern slavery in supply chains of global fashion brands	4 years	1	MS15
	Sustainability Officer	A global omni-channel retailer overseeing modern slavery in supply chains of global fashion brands	3 years	1	MS16
	Group Ethical Sourcing Manager	A global food retailer overseeing modern slavery in food supply chains	3 years	1	MS19
	Senior Human Rights Manager	A global food retailer overseeing modern slavery in food supply chains	5 years	1	MS20
Service provider – accreditation, certification body	President & CEO	Providing governance over auditors' competence; a global outreach with insights into auditors' competence and compliance	6 years	1	MS9
	Founder and Director	Primarily an auditing firm assisting clients to develop compliance mechanisms	3 years	1	MS8
	Social Project Manager	A global governing body with insights into the performance of leading certifiers across 70 plus countries	1 year	1	MS13
	Strategic Development Director	Leader of a team of global assessors that oversee a network of actors operating in various industries across 70 + countries	14 years	1	MS17
	Manager	A global actor providing education, advocacy, and third-party assurances in the seafood sector	5 years	1	MS31
Service provider - technologies	Freelance Auditor	An auditor with extensive experience with a global governance body, currently working as an auditor and auditor rights advocate	15 years	1	MS18
	Head of Social Research & UK Operations	A data platform provider working with multiple supply chain actors that share data and insights	4 years	1	MS12
	Director of Programs	A multi-channel mobile platform for engaging actors and measuring risks; working with energy, mining, agribusiness, and others	8 years	2	MS3, MS7
	Senior Sustainability Expert	A data platform provider working with multiple supply chain actors that share data and insights	3 years	1	MS23
Service provider – consultancy, training & research	Vice President Asia Pacific	Provides research, training, consulting, and policy advocacy across a variety of sectors globally	9 years	1	MS24
	Senior Program Director	Provides research, training, consulting, and policy advocacy across a variety of sectors globally in over 70 countries	17 years	1	MS26
	Executive Director and Founder	Focuses on loopholes in domestic and international legal regimes to hold companies accountable for abuses in their global supply chains	6 years	1	MS27
	Senior Advisor	A group of 70 plus organizations providing research, advisory and training services focusing on global health, economic development, and gender	6 years	1	MS28
	Executive Director	A director of a research centre leading research in modern slavery (with previous experience with a global regulatory body)	3 years	1	MS29

directors in our sample, those roles were often preceded by on-the-ground experience in auditing, consultation, and advocacy on modern slavery. Importantly, our respondents were also involved in oversight of a large number of actors and across a wide variety of sectors (e.g., agriculture, seafood, manufacturing). While these respondents were not able to share any confidential data with us, their experience positions them well to answer broader questions about collaborations and technology adoption. We believe this to be a strength of our data – our experts had experiences across the different roles and organizations involved in modern slavery. Therefore, while we have categorized the respondents by five actor groups in Table 3, this is based on their roles at the time of the study. Many of the respondents could have been categorized into other groups at different points in their careers.

The main source of data for this study were 32 semi-structured interviews conducted between August and December 2021. A recent report by Gartner provided background on actors involved in modern slavery (Cossio and McNeill, 2021). Initially, we developed a list of 10 global NGOs and initiated our interviews with them. We aimed to capture a variety of perspectives and were guided by referrals from them to increase and diversify our pool of respondents. As noted, all of the experts were deeply involved in modern slavery and had served in many roles to help address it, such as managing supply chains, technology providers, advocacy groups, victim support, and leading multi-actor initiatives.

The interviews lasted an hour each on average and were conducted over Zoom. All interviews were recorded and transcribed verbatim. During the interviews, we were guided by a set of open-ended questions. We probed the respondents if needed – keeping the research question in mind. For example, we asked “Can you highlight some innovative practices that help detect modern slavery in firm’s supply chains?”, “Is there any opportunity for ongoing use of technologies in firm audits?”, and “What are some of the technologies that you use to help give you visibility and assurance in the company supply chain?” We followed-up with probing questions such as “How do you triangulate the data?”, “How have different technologies helped provide greater visibility and transparency from your organization’s perspective?”, and “Do worker voice technologies lead to the identification of modern slavery?” We also explicitly asked all respondents about the roles played by the different actors in identifying modern slavery.

3.3. Data analysis

Broadly, we started the analysis with our research question. We also grounded our analysis in the literature and theory. As shown in Table 4, we observed the existence of two domains identified in the literature (Gold et al., 2015; Simpson et al., 2021), i.e., the contracts- and conditions-focused domains. Consequently, in the analysis, we focused on the differences in how organizations cooperate in these domains and how ‘domain membership’ shapes the collaboration. As discussed earlier in the paper, we were also guided by RDT, with an emphasis on mutual dependencies.

We then performed two key rounds of data analysis. The data analysis followed procedures that are commonly adopted in qualitative research, including best practice for using quotes to present claims (Rockmann and Vough, 2024). We did not have any *a priori* coding scheme and “let the data speak” (Huxham and Vangen, 2000). Whilst performing the first round of scanning through the data we focused on developing broader understanding of (a) what collaborations organizations pursue and why; (b) how technologies are used in collaborations; and (c) how the unique context affects the collaborations and use of monitoring technologies to identify modern slavery. After performing the first inspection of the data, we identified high-level patterns linked to those three questions (see Table 4). We compared our findings (the authors conducted the coding independently) and, as shown in Table 4, grouped the patterns into preliminary themes. In the second round of analysis, we focused on establishing the linkages between the

Table 4
Overview of data analysis.

Points of departure	Preliminary themes	Contingencies
Research Question How can multi-actor collaborations leverage monitoring technologies to identify modern slavery in global supply chains? Theoretical underpinnings There are two domains of modern slavery: contracts- and conditions-focused. RDT’s mutual dependence perspective helps explain when and why different organizations collaborate with each other. There are inter-dependencies between collaborations and the use of monitoring technologies to identify modern slavery.	Collaboration motivation The motivations to collaborate underpin expectations on how modern slavery activities are organized, including which actors are engaged. Modern slavery domains The two domains of modern slavery (i.e., contracts- and conditions-focused domains) each have their own <i>modus operandi</i> , which impacts approaches to identifying modern slavery. Collaboration modes There are two levels of collaborations to identify modern slavery across the two domains: micro- and macro-levels. Trust building Trust is a foundation of close collaboration (e.g., data collection, exchange, and analytics) within and across the two levels. Boundary spanning activities The two domains and levels are co-existing and connected; they do not operate in isolation. Activity alignment To leverage technology through collaborations, actors must recognize mutual dependencies in aligning their activities.	Modern slavery posture This is characterized by the contracts- and conditions- focused domains. Collaboration scope This is characterized by micro- and macro-level collaborations. Cross-boundary interactions This is characterized by collaborations between different motivation postures and different collaboration scopes.
	Resource access Identifying modern slavery in different contexts requires access to different combinations of resources held by different actors. Context matching Resources, including technologies, must be matched to the unique context and collaborations in any supply chain, considering potential adverse effects.	Contextual embeddedness This is characterized by tailored approaches to identifying modern slavery grounded in collaborations between key actors.

preliminary themes and the outcome of identifying modern slavery. As part of this process, we identified four key contingencies. The contingencies formed the basis for the conceptual framework outlined in Section 5.1.

4. Findings

This section presents the results of our data analysis. Building on the data analysis procedures described in Section 3, we present the findings

around our four key contingencies, focusing on how they facilitate (or inhibit) identifying modern slavery by leveraging monitoring technologies through multi-actor collaboration. Collectively, the findings provide a basis for directly answering our research question.

4.1. Modern slavery posture

Modern slavery posture refers to the reason why an organization pursues collaboration. There are many reasons why various actors in supply chains do not want modern slavery to be found. For example, it can be highly profitable for those exploiting others. However, they could also be severely punished if it is located, such as reputational risk for a firm, or criminal liability for an individual. Motivation is thus impactful in determining the modern slavery posture of any given actor – a firm, auditing body or NGO. This is illustrated by the following quote:

“In terms of how difficult is it to find, it is. It depends on whether you’re looking for it and it depends on whether you want to find it.” MS4

The contracts- and conditions-focused domains provide contrasting environments in which modern slavery is identified. We note that many interviewees indicated the existence of these domains, though the terminology used to describe them varied. For example:

“I think those are kind of the two camps that I’ve noticed. One kind tends to be used to cater to corporations’ risk-based approach and then the NGOs tend to come in for more continuous improvement approach. Or not necessarily always NGOs, they could be consultancies, they could be social enterprises, but I think there’s a bit more potentially, a more holistic approach to looking at the root cause rather than a compliance-based approach.” MS7

In the contracts-focused domain, firms seek collaborations and secure resources predominantly to acquire compliance with standards and other reference points. In fact, respondents frequently referred to this domain as the “compliance-domain” in the interviews. Firms are increasingly pushed to address modern slavery through audits, certifications, and other compliance mechanisms and look for actors that can provide such services. We also heard many comments about audits focused on confirming the absence of modern slavery, in many cases as a mechanism to avoid trouble. To illustrate,

“I think that the compliance-based approach is very familiar to companies. So, I think many of them still seek social audits as from a compliance-based perspective and a risk mitigation, risk minimization perspective. But there is definitely a trend now and a lot of spotlight on companies that move away from trouble, some suppliers or regions or commodities, rather than actually working with the supply chain to improve it.” MS7

The contracts domain is thus characterized by actors whose primary interest is in confirming the absence of modern slavery in their supply chains, not necessarily identifying (and remediating) it. Moreover, this domain is dominated by firms and service providers. Although other actors might get involved (e.g., an NGO might assist with data collection), this involvement tends to be limited or reduced to symbolic collaborations. Notably, and consistent with RDT, the firms and service providers have significant control and discretion over the allocation of resources and their use. For example, the audits (and related data) have a commercial value, which is protected and leveraged for the benefit of participating actors (i.e., a certificate demonstrates absence of modern slavery). Actors in the contracts-focused domain are primarily concerned with improvements of the audit process, such as cost-cutting or audit integration.

In the conditions-focused domain, actors are interested in identifying the presence of modern slavery (and then working to remediate it). This domain is dominated by NGOs, labour unions, and other actors whose *raison d’être* is different from firms seeking compliance and

certification. These actors also employ different mechanisms to identify modern slavery: there is less focus on audits and more focus on investigative work and working with local communities as the following quote illustrates:

“NGOs are going to identify [modern slavery] better, but also it’s not just sort of putting a red dot on a map to say there is forced labour there. An NGO is going to do a better job of working with a local community and identifying what the appropriate remedy is. Because, in the end what’s going to prevent forced labour in a particular industry or what’s going to stop existing forced labour or what’s going to remedy forced labour is kind of context specific.” MS27

Recognizing the two modern slavery domains has important ramifications for selecting collaborating partners since each domain has different aims and approaches to identifying modern slavery. From a RDT perspective, for instance, it means that actors seek to secure different resources and align (or naturally drift toward) a domain that provides such resources and which allows the actors control over the allocation of resources for their intended purpose. An example might be a firm that seeks to legitimize practices in their supply chains (but not necessarily improve these practices), is likely to collaborate with actors that provide such service. Consequently, the selection and adaptation of technologies also serves as a tool to leverage the legitimization of supply chain practices. For example, worker voice technologies (WVT) can be used to actively conceal labour issues by careful selection of participants or through persecution of whistle blowers. Hotlines can be similarly used to monitor the workforce and identify workers the company may view as problematic. Conversely, technologies supporting more genuine motivations can have the opposite effect.

The adoption of specific monitoring technologies is driven by the objectives of the collaboration. For example, an organization may seek improvements to its data collection process to achieve greater audit efficiency – a typical approach for actors operating in the contracts focused domain. Actors involved in audits can benefit from the digitalization of the audit process enabled by various technologies (e.g., data collection platforms; phone-based apps, etc.). As the following quote illustrates, this can enhance audit efficiency:

“I think technology in the audit process, I mean certainly one thing that’s helped is for the auditors to be able to answer the questions. On their audit protocol for them to be able to go through and answer those questions you know kind of in real time. And you know just save them time and energy and also as soon as you’re writing a piece of paper and transferring it. [...] The fact that the reports get automatically uploaded to [audit platform X] or [audit platform Y] or whomever I think that’s also really helpful ...” MS9

4.2. Collaboration scope

We found two principal modes of collaborations: *micro-level collaborations* enable identifying modern slavery at a facility level, whilst *macro-level collaborations* enable the same at a regional or national level. In both domains, collaborations to secure resources and capabilities emerge at these two levels. In the contracts-focused domain, collaborations emerge to enhance the efficiency of compliance mechanisms, such as audits. In the conditions-focused domain, the focus was on mobilizing communities to identify modern slavery (see Table 5). Cross boundary collaboration is also possible, as we explore in Section 4.3.

Collaborations engaged in by each actor depends on the modern slavery posture. For example, audits performed at individual facilities could be used for multiple purposes – an activity typical for the contracts-focused domain. This is a focus of so-called audit platforms – data collection and exchange mechanisms, which are developed to enable an exchange of insights gained through an audit at one facility to inform further insights for various supply chain actors, who use the same facility. If actors collaborate across the micro- and macro-levels, audit

Table 5
Collaborations for identifying modern slavery.

	<i>Contracts-focused domain</i>	<i>Conditions-focused domain</i>
Micro-level collaborations	Collaborations to enable audits at facilities PRIMARY FOCUS: To ensure compliance against a standard USE OF TECHNOLOGIES: Monitoring technologies enable data collection for auditing purposes (e.g., sensors)	Collaborations to interlink local communities and workers PRIMARY FOCUS: To mobilize (and give voice to) local communities to assist in revealing modern slavery USE OF TECHNOLOGIES: Monitoring technologies enable data collection from underrepresented communities (e.g., worker voice technologies)
Macro-level collaborations	Collaborations to enable supply chain audits PRIMARY FOCUS: To ensure efficiencies in supplier auditing confirming the absence of modern slavery USE OF TECHNOLOGIES: Firms can enhance data collection with data exchange technologies (e.g., cloud-based applications)	Collaborations to interlink local communities across regions and countries PRIMARY FOCUS: To ensure effectiveness of community mobilization in sharing insights across the region or country to assist in identification of modern slavery USE OF TECHNOLOGIES: Firms can use macro level data and analytics for identifying hot spots (e.g., satellites)

results and other data can be shared to explicitly serve the interests of multiple actors. This means that audit requirements can reflect the needs of multiple actors; an integrated audit can therefore cover areas that would normally require multiple audits, thus contributing to audit optimization. The connection between the modern slavery posture, selection of a collaborator, and choice of monitoring technology is highlighted by the following quote:

“So, where we’ve partnered with people, like [software platform], They had a very similar mindset to us. We set up us originally because we recognized that there were so many different platforms, so many different certifications, everything like that. And then suppliers are just exhausted because they’re having to answer everything separately, I mean that’s a full-time job for a few people.” MS12

The collaborations in the conditions-focused domain tend to involve multiple actors who collaborate for a ‘greater good’. These collaborations, which underscore mutual dependencies in building capacity and social capital in service of the larger goal of identifying modern slavery, can also occur at both the micro- and macro-levels. At a micro-level, these collaborations aim to interlink local and other actors. To provide one example that was raised several times in our interviews, an NGO may work with local communities to gain on-the-ground insights on the labour conditions of workers. At a macro-level, the actors collaborate to identify modern slavery in a particular location, but also work together to collect insights across regions and identify hotspots of modern slavery. If hotspots are identified in a particular area, the insights can be used for local actors to follow-up and investigate. Importantly, these insights are rarely developed by any one actor on their own, further underscoring mutual dependencies.

“The technology has enabled not only to facilitate expedited operation of the hotline, but also [to] connect the resources through these different platforms of Salesforce and volunteers and do this heat map, where the situation and geospatial location, etc., [...] We analyze all these different sorts of information that we’re gathering from the interaction with survivors of trafficking to sort of identify certain trends, when [the] same addresses and information about trafficking comes [up] over and over so, then we can eventually

develop customized packages for law enforcement about information about traffickers, never about victims because [of] confidentiality.” MS11

The collaborations across micro- and macro-levels enable exchange of insights and data for mutual benefit. Technology can enable the exchange of insights (such as data or results of analytics). Whilst the free flow of data is essential for collaborations, the flow of data is often difficult to achieve. To start with, standardization of data collection is critical, and this will be based on the need to coordinate and collaborate with many actors. Many respondents explained that the lack of data format standards and standardized data collection processes hampered effective data exchange between actors:

“So one of the kind of main criticisms of the social audit world is lack of standardization in the questions that are being asked and the data that’s being collected. [...] It’s really, really hard then to kind of amalgamate that data together and make comparisons if everybody’s looking at different things at different times ...” MS6

In part, the exchange of data is influenced by a willingness of individual actors to collaborate. This willingness is often grounded in the trust, or lack thereof, between potential collaborators. For example, complicated agreements over who owns the data may prevent collaboration. An illustration is whether service providers of worker voice technologies (WVT) own the collected data. Depending on any particular agreement, they may collaborate on projects where they are in control of the data or outsource their technology to actors who own (yet not further share) the data.

Other monitoring technological solutions may have fewer proprietary restrictions. For example, satellites are increasingly being used to monitor large geographic areas (e.g., forests, shrimp farms, food processing facilities). While not necessarily definitive, this type of data can highlight potential risks requiring further investigation, mindful of the cost of acquiring this data. The benefit of using it is that local actors could be alerted to verify any concerns, as the following explains:

“We have a couple of projects where we’re looking at using satellite imagery of monitoring forests. [...] There’s some kind of the satellite spots deforestation, a quick survey goes out to a group of, you know, at local community members asking them, hey, what’s going on? Can you find out? That kind of triangulation of data I think is kind of the future of integrating these technologies together because if we’re all collecting information in silos and not sharing, we’re not really leveraging.” MS7

Technology can also enable firms to push data collection deep into their supply chains, including well beyond the first tier. This requires technological interoperability, but it also requires trust throughout the chain on how the data will be collected and used, particularly given the sensitivities around an issue like modern slavery. Financial and informational resources must thus be supported by trust, legitimacy, and an understanding of local contexts. In cases where trust-building is at an early stage, monitoring technology can enable anonymous reporting, which could compromise identification of issues at the micro-level, but might enable improved analytics at the macro-level – as illustrated by the following quote:

“So it’s the lower tiers where most of the risk is because the funding and the work isn’t getting down that far. But if I was a tier one supplier and I don’t want to be cut out with the supply chain, how do I share that information with my brands at all? I think that allowing anonymous reporting is important so that you can at least find the risks and then figure out where to go for those suppliers that are particularly edgy about working with the suppliers to capacity building [...] But then letting the suppliers have hold of that data, you know in our system if it’s multi tier. The tier one supplier gets to crunch all of the data and run analytics on their suppliers, and it trickles down the chain. They can look at what’s going on with the

people they're working with and set corrective actions and help to improve things. So, I think it's more about having agency and ownership of projects to enable suppliers to feel confident in participating." MS12

Data collection is ultimately underpinned by trust and is only as effective if it is transparent and empowers workers to safely share their views. In the absence of such conditions, workers may not participate or may provide biased feedback, thus undermining efforts to identify modern slavery.

"If you're collecting people's data and you are collecting very sensitive information about the labour conditions, you're asking them to make complaints or you are recording grievances that they've raised, if that information gets into the wrong hands, gets into the hands of the very people that are abusing these people or you know, gets into the wrong hands in any other way, that can then create a risk for that person, and this is one of, this is one of the kind of big reasons why some worker voice apps really struggled to, to take off because workers are nervous to, to use them [...] That risk may even be further perpetuated if you're talking about ethnic minorities, migrant workers, or even kind of illegal migrants, who don't want to be identified ..." MS6

Without confidence their data is protected, workers may not report on issues such as if they are regularly paid on-time, if they paid a recruitment fee, or if they retain access to their personal identification documents.

4.3. Cross-boundary interactions

Notwithstanding the distinctions between the domains, the contracts- and conditions-focused domains are neither disconnected nor operate in isolation. The domains are co-existing, and both can contribute to identifying modern slavery in supply chains over time. We found several indications that collaborations emerge across these seemingly separate domains, particularly when the collaborations emphasize identifying modern slavery, rather than focusing on achieving aims congruent with their domains (e.g., legitimization of modern slavery initiatives through auditing and certification as observed in the contracts-focused domain).

We found evidence that *cross-boundary interactions* can leverage domain-specific resources and provide unique opportunities for collaborations for actors from each domain. For example, for firms a collaboration with NGOs or other actors from the conditions-domain can lead to increased effectiveness of audits – arguably leading to increased legitimacy. Facility audits are good examples of such collaboration as they can have serious limitations. In many cases, audits are limited to the grounds or specific activities of a facility (typical in a contracts-focused approach), but may benefit from exploring linkages to external resources held by the broader community and seek insights from multiple sources of data (typical in a conditions-focused approach). Facility audits are usually announced well in advance, meaning a facility manager can 'set' the scene. If the managers/owners purposefully conceal evidence, it could be extremely difficult for an auditor to identify modern slavery. For example:

"We found by very, very conservative estimates that a third of migrant workers were enforced labour. Whereas we found that audits found that less than 1 % of workers were forced labour. I think that the difference there is once audits are on site; they're pre-announced, there may be worker coaching." MS26

These are serious vulnerabilities, but the effectiveness of audits improves through collaboration with actors having expertise and resources within the conditions-focused domain. Such actors (e.g., local NGOs or communities) may not necessarily be aware of the specific compliance mechanisms an auditor follows. However, they can provide broader

insights into the working conditions at a facility that can inform an auditor about 'hidden' problems. Such collaborations provide access to critical resources, such as 'informal' channels and first-hand data, that is typically difficult to obtain. They also highlight the unique resources different actors possess and the mutual dependencies in high veracity audits.

Boundary spanning activities can help an auditor to focus on matters that are important, but not necessarily directly observable. However, boundary spanning activities require a clear mandate and resources from a firm or the auditing organization to enable auditors to engage in them. We found that some firms engage in these activities, yet there are still several conditions, such as the modern slavery posture of the firm and the experience of the auditor. As one interviewee explained:

"Experienced auditors are typically OK. They will check the background, we will get the information from the previous audit or what is happening, a Google search in the area, see if there is any issue with that particular facility, then he will go in. [An] inexperienced auditor will jump into the facility [unprepared] and he will say that 'everything is okay'." MS18

Cross-boundary interactions, however, must be supported by a greater range of resources than activities confined to one domain alone. For example, auditors conducting a boundary-spanning audit will require a wider set of skills, knowledge, and experience to gather, analyze, and interpret the broader types of evidence provided in each domain. Gathering boundary-spanning evidence will, in turn, require increased financing to support the broader scope of activities, access to people in a wider range of contexts (e.g., facilities, communities), and the legitimacy that enables such access. Finding auditors with the ability to conduct boundary spanning audits can be challenging, however, and points to the need for new forms of auditor training. As one expert remarked:

"There are social compliance auditors out there and they're trained to look for social non-compliance. But a lot of the auditors who are out there might have a technical background. [...] They can see if there's live wires or slippery falls and things like that. That's physical; that's tangible; but forced labour indicators are non-tangible. They find it incredibly hard to detect and the interview techniques that you need to use to detect these things when you go out and actually speak with management and workers are so different from what they're used to. [...] Building capacity [is] a really long and slow process and incredibly frustrating." MS25

We also found examples of firms operating in both domains. For example, a large seafood firm engaged in a contracts-focused approach through social compliance auditing, but also hired local NGOs to investigate labour hiring practices and observe the well-being of workers (especially migrant workers), indicating a conditions-focused approach. Sharing the results from one approach potentially benefits the other. Other actors, such as NGOs, can operate in either domain, but can also explicitly focus on facilitating knowledge transfer from one domain to another.

Collaborating across boundaries also has ramifications for how actors leverage monitoring technologies to improve the likelihood of identifying modern slavery. Here, data analytics provides great promise in identifying modern slavery, though its use in that context remains in the early stages. For data analytics to work well, however, it depends on the data collection and data exchange and their integration into the data analytics process. Data analytics is not only about crunching data; it is about using and exchanging the results between actors to facilitate the flow of information. This underscores the importance of aligning activities across collaborators and domains. As one respondent asserted:

"Technology has a role to play, but for us is more like how you implement flows of information with workers and about the workers both ways. So then, it's about transparency [...] but you first need to

create that environment where flows of information come and go for them all the time.” MS11

As part of this process with clear mutual dependencies, boundary spanning activities play an important role. For example, in the context of WVT, collecting the data outside of the workplace, such as with the help of a trusted local NGO, illustrates how collaboration among diverse actors can improve the likelihood of identifying modern slavery. Building a database can then provide a basis for a data analytics program to indicate potential hotspots. As one respondent explained,

“[A platform] doesn’t give you a definitive “this person is in modern slavery” but rather it presents you with, okay, these are the various risk factors here. [...] This person doesn’t have access to their ID documents. They weren’t paid on time last month. They did pay a recruitment fee to get this job and then you’re able to build up a bit of a picture and you as an auditor or a frontline responder can say, hey, I think there might be a modern slavery situation here.” MS6

The analytics are only as good as the input data, and collaboration is therefore a key enabler of data analytics. Actors collaborate to secure access to valuable resources, whether they are technological or administrative, to identify modern slavery. Critically, as the evidence demonstrated, these resources can be held by different actors from both domains. Technology providers, for example, may either hold or can facilitate access to key worker response data. However, it may only be possible to collect this data by drawing on the resources of other critical actors, such as contextually embedded NGOs holding local knowledge, access, and trust. Data collection, data exchange, and data analytics are therefore all grounded in mutual dependencies requiring good-faith collaborations focused on aligning key activities.

4.4. Contextual embeddedness

Context is also important for identifying modern slavery as different approaches will be more applicable in different situations. One participant went so far as to suggest that identifying modern slavery in practice means “you’re trying to figure it out on a case by case basis” (MS27), while another noted that you must put what you know “within the context of that universe” (MS23). Recognizing the context is critical and a failure to do so hampers the ability to identify modern slavery. For example, an often-cited example focused on auditor skills as one participant explained:

“It’s been quite interesting to see the difference in approach and sometimes the auditors are lacking that nuance and really kind of the reality of context on the ground to put some of the information to analyze it against that.” MS7

Resource access is one of the keys to tailoring approaches to identify modern slavery to the specific context of any one supply chain. Resource access is often underpinned through collaborations. For example, one key resource is physical access to local environments. Physical access must also be supported by the resources needed to identify modern slavery once on-site, such as language skills, perceptions of legitimacy, and technology. As our collected evidence made clear, understanding the context (in a facility or, more broadly, in a geographical location) assists collaborating actors in developing their strategies for identifying modern slavery. The following quote illustrates the importance of contextually embedded NGOs in providing access to resources required in many contexts, such as local translators and, more importantly, local networks and worker trust:

“You know what we draw NGOs for? We draw on them for an advice that is localized in the different regions we work and who to align with if we have a case to remediate or there is a victim. So really leaned on people who have some expertise on the ground where we source from through our supply chain, who understand the real issues in our supply chain.” MS15

Building on the point above, the need to understand the cultural and linguistic context of the workers was repeatedly emphasized. Collaborations are critical in providing an understanding of the needed context. One expert explained the potential roles of NGOs in doing so:

“One is that for solving a problem you need to understand the problem, so if you need to create a works and providing the information doing research on understanding the context, so what is not working, why is not working is the first step so certainly NGOs could play a pivotal role on that providing background context and perspectives.” MS14

Understanding the broader community context in which modern slavery occurs is also essential. Auditors can reach out to local communities (i.e., not only the workers themselves) with the help of local NGOs. This can be for the purpose of connecting with the broader community, but also for providing an additional layer of safety for the auditor. Improved safety, for example, could potentially be enabled through wearable technologies, such as having an on-the-ground community actor monitoring the location of the auditor while they were on-site. One participant explained the importance of connecting to the broader community:

“I think, which has also been innovative, is really attempting to the extent possible to interview workers in their communities rather than at their workplaces or in areas outside of their workplaces. So you know, on Sundays workers that work six days a week, they might go to the market for a couple of hours, so we are able to interview them there, sort of offsite at worker housing.” MS26

Contextual embeddedness also means matching the monitoring technologies with the environment. Sophisticated technologies are not necessarily needed; basic technologies (such as mobile phones, cameras, or hotlines) can also be effective and appropriate in different situations. Technological solutions often need to be agile and suitable for the low-technology environments in which modern slavery often flourishes. Moreover, sophisticated and basic monitoring technologies can be used together to increase the likelihood of identifying modern slavery. This is illustrated in the following quote:

“So, in certain factories, workers have high levels of literacy and access to smartphones, an app-based, text-based grievance mechanism might work great [...] In the coffee sector with funding from [redacted] on a needs assessment [we] found that the vast majority of workers didn’t have a smartphone, didn’t have Internet access, very low levels of the literacy, [and were] non-Spanish speakers. In that case, we actually determined that the lowest tech solution was the most appropriate that workers prefer. We also asked about their preferences, they preferred face to face interaction. Telephone calls also played apart and community radio and announcements, for example, but realizing that context, a high tech solution really wasn’t going to be appropriate to workers needs and technology access.” MS26

Any approach to identifying modern slavery must be mindful of the potential adverse effects in different contexts. One risk in some environments is that there can be a dark side to technology. For example, some interviewees noted that the technology might be used to exert control over workers. There are also many challenges in implementing technologies in practice, such as ensuring poor data are not codified, interoperability, data security, and the high cost of mobile data, to name just a few of the illustrations provided by respondents. These challenges can vary between supply chains, so it is important to be mindful of unique situational contingencies. Yet, all of the experts also agreed that monitoring technologies can play an important role in the process of identifying modern slavery if it is leveraged through collaborations between key actors in the system.

5. Discussion

Our study provides several contributions, including further evidence on the existence of the contracts- and conditions-focused domains, as well as extending understanding of those domains with evidence of micro- and macro-levels of collaborations and boundary spanning collaborations. Importantly, our study also identified four key contingencies facilitating or inhibiting the identification of modern slavery by leveraging monitoring technologies through collaborations. In this section, we discuss a conceptual framework grounded in our analysis and discuss the contribution of our work to the literature.

5.1. Conceptual framework

Our findings provide the foundation for developing a conceptual framework for leveraging technology through collaboration to identify modern slavery (Fig. 1). Our framework explains how multi-actor collaborations can leverage monitoring technologies to identify modern slavery – facilitated (or inhibited) by four key contingencies (i.e., modern slavery posture, collaboration scope, cross-boundary interactions, and contextual embeddedness).

As shown in Fig. 1, the framework recognizes the contracts-focused and conditions-focused domains. In the contracts-focused domain, multiple actors develop collaborations and exchange resources to further their strategic goal of confirming the absence of modern slavery. The collaborations and resources required to confirm the absence of modern slavery are based in the modern slavery posture of the actors. The collaboration scope can, therefore, vary from one supply chain to another, potentially addressing collaborations at the micro- and macro-levels. Grounded in RDT, these collaborations will be focused on ensuring the necessary resources are available to confirm the absence of modern slavery. In other words, the actors seek collaborations that address resource needs congruent with their strategic goals. As actors' motivation to identify modern slavery increases, collaborations have the potential to broaden in scope at the micro- and macro-levels to ensure that the needed resources are available to address those needs. This exchange of resources includes technological solutions, such as worker voice technologies, hotlines, or exchange of data through auditing platforms. The contracts-focused domain, therefore, demonstrates the linkage between modern slavery posture, collaboration scope, and efficiently leveraging monitoring technologies through collaboration to confirm the absence of modern slavery in a supply chain.

The conditions-focused domain has the strategic goal of confirming the presence (where it exists) of modern slavery. This domain also originates from modern slavery posture, which is grounded in an actor's purposeful decision to explicitly identify modern slavery. Recognizing modern slavery posture as a starting point for both domains underscores the potential for cross-boundary interactions. The framework, therefore,

recognizes that the contracts- and conditions-focused domains are co-existing and can both contribute to identifying modern slavery. The framework also recognizes that collaborations can bring mutually dependent resources (such as trusted local informants) to bear through both domains, as well as across the micro- and macro-levels. Moreover, the conditions-focused domain requires that cross-boundary collaboration is tailored to the unique supply chain context. Identifying the presence of modern slavery, particularly in cases where it is concealed, requires an approach specifically applicable to the context of the unique circumstances of that supply chain. These tailored approaches can include monitoring technologies leveraged through multi-actor collaboration. Starting from modern slavery posture, this domain therefore demonstrates the linkage between cross-boundary interactions, contextual embeddedness and effectively leveraging monitoring technologies through collaboration to confirm the presence of modern slavery (where it exists) in supply chains.

5.2. Contributions to the literature and theory

Our study supports the literature showing that collaborations are driven by a pursuit of strategic goals through recognizing and developing mutual dependencies (e.g., Berkowitz et al., 2020; Kretschmer et al., 2022). In the modern slavery literature, this pursuit is commensurate with identifying modern slavery in the contracts-focused and conditions-focused domains (Simpson et al., 2021), with the former associated with confirming the absence of modern slavery, and the latter associated with confirming the presence of modern slavery. Our study also shows that RDT is a valuable outcomes focused approach (see Oliveira et al., 2022), which helps to explain the advantage provided by collaboration among different key actors to identify modern slavery. Technology adoption in collaborations is also influenced by the strategic goals of collaborating actors. Consequently, collaborations based around technologies can advance positive outcomes, such as increased veracity and timeliness of monitoring activities in supply chains (Castka et al., 2020; Reid and Castka, 2023), but also potentially lead to concealment and legitimization of illegal or unethical activities, such as increased control over the workforce (Mol, 2010). While previous studies have alluded to some aspects of collaboration in the context of modern slavery (Gold et al., 2015) or investigated the adoption of technological solutions in this context (Marques et al., 2024), this research widens our understanding about strategic actions of actors through collaborations, with a specific focus on monitoring technologies (see Table 6).

At a more nuanced level, we expand the literature by demonstrating how four key contingencies (i.e., modern slavery posture, collaboration scope, cross-boundary interactions, and contextual embeddedness) facilitate (or inhibit) leveraging monitoring technologies embedded in collaborations. First, our evidence reveals that the posture taken by different actors will shape why collaborations occur and the type of

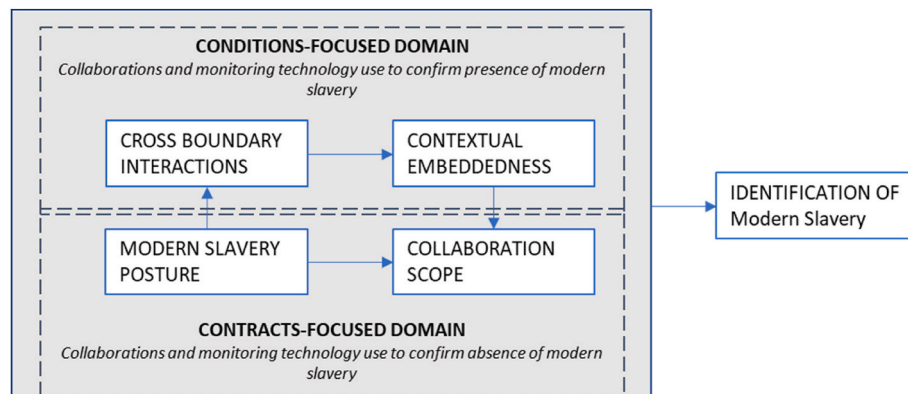


Fig. 1. Conceptual framework for leveraging monitoring technology through collaboration to identify modern slavery.

Table 6
Summary of findings.

Contingencies				
Use of technologies	Modern slavery posture	Collaboration Scope	Cross Boundary Interactions	Contextual Embeddedness
	Selected technologies reflect the modern slavery posture (i.e., technologies in the contracts-focused domain are intended to confirm the absence of modern slavery, while technologies in the conditions-focused domain are intended to identify it). Securing needed resources to establish and/or maintain capabilities aligned within the modern slavery posture	Increased scope of collaboration broadens the use of technologies – characterized by a shift from predominant use of technologies for data collection to data exchange and analysis Securing needed resources to develop capabilities within the modern slavery posture	Cross boundary interactions can facilitate a shift in the use of technologies to serve different ends Securing needed resources to repurpose the modern slavery posture	Contextual embeddedness facilitates the veracity in the use of technologies, primarily for data collection Securing needed resources to deepen capabilities

technologies most likely to be adopted. The desire to either show the absence (contracts-focused domain) or presence (conditions-focused domain) of modern slavery in supply chains is an important contingency in the nature of multi-actor collaborations. Specifically, our study shows that different actors will engage in collaborations for the purpose of acquiring different valued or needed resources to satisfy either domain. In turn, this will influence actors' assessment, and subsequent use, of different monitoring technologies.

Second, there are implications for the scope of collaborations and whether this occurs at the micro- or macro-level. While both are important, the former focuses on the facility level with the latter addressing the regional and/or national levels. As monitoring technologies are resource intensive (i.e., satellite imagery and its analytics is not a core competence of many actors) and not central to many firms' *raison d'être* (i.e., monitoring of working conditions is not a primary focus of labour intensive supply chains unlike, for instance, quality management and focus on efficiency of supply chain operations), actors seek collaborations to gain benefits from technologically driven solutions across both levels. While the pursuit of increased collaboration scope is often driven by the intrinsic motivation of different actors, it is notable that actors holding unique competencies at the micro- and/or macro-level play increasingly important roles in identifying modern slavery. Arguably, the extended scope of collaborations is driven by competencies that are enhanced by technologies. It may be argued that prior to the digitalization of processes in supply chain audits and the more ascendant role played by big data and AI analytics, there was comparatively less reason to develop such collaborations.

Further, our research supports earlier studies (Fawcett et al., 2008; Sodhi et al., 2022; Marques et al., 2024) arguing that social capital, such as trust and willingness to collaborate, are essential to applying technologies in a supply chain context, in our case to identify modern slavery. However, we also demonstrate that actors develop social capital to advance their organization centric benefits (contracts-focused domain), but also for a 'greater good' (conditions-focused domain). Our theoretical contribution here is to show that RDT can be usefully adopted to focus on other motivations of actors besides financial performance (or firm survival). Such mutual dependencies can also help fulfil societal norms and expectations around the identification of modern slavery in global supply chains. We contend that RDT is a useful line of inquiry in modern slavery and supply chain research; it offers scholars a different explanatory framework in addition to theory more commonly used such as institutional theory and global value chain theory (Strand et al., 2024, p. 6).

Third, as actors explore more ambitious goals through developing cross-boundary interactions (such as data exchange and its multipurpose utilization), long term relationships, along with trust and enhanced capabilities, become a more central focus in collaborations. Such developments can further leverage the potential of technologies for data collection, exchange, and analytics. Our argument, and consistent with RDT, is that engagement with each of these features (i.e., data collection, exchange, and analytics) will depend on what purposes they individually and collectively serve in meeting different actors' resource needs.

Fourth, the need to acquire and manage important resource dependencies also enables collaborations of actors from contrasting domains, as well as facilitates contextual embeddedness. The extant literature often identifies contrasting domains, such as symbolic versus substantial domains mentioned earlier (Prajogo et al., 2021), but rarely theorizes how the contrasting domains interact. We demonstrate that resource dependencies allow for collaborations across domains but must remain mindful of different settings. This argument urges some caution in assuming that the identification of modern slavery through monitoring technology in multi-actor collaborations will be the same across different contexts. The success or otherwise will depend on aligning suitable monitoring technologies to the specific environment, the needs of different actors including their modern slavery posture, as well as the scope of the collaboration.

Previous studies have demonstrated that actors tend to focus on developing collaborations that assist them in advancing their strategic goals (Shiu et al., 2023); for example by developing dyadic relationships with individual actors such as certifiers (Castka et al., 2015). Organizations operating in such institutional environments thus primarily seek to maintain their autonomy (Drees and Heugens, 2013). In the context of modern slavery, the autonomy is, in part, preserved by membership in a particular institutional domain (i.e., contracts-focused domain). Our study has extended understanding of the contracts-focused and conditions-focused domains (i.e. Simpson et al., 2021), establishing the conditions under which actors are more likely to leverage monitoring technologies and develop mutual dependencies commensurate with relevant strategic goals. We also identified micro- and macro-levels of collaboration as well as collaborations within and between the domains (see Table 5).

The common perspective of RDT is to focus on the performance of the dependent (focal) organization, as earlier reviews and meta-analyses have widely observed (Drees and Heugens, 2013; Hillman et al., 2009; Ozturk, 2021). Our study of collaboration and technology extends the focus of RDT to a multi-actor context, while providing theoretical contributions to the mutual dependence perspective of RDT. RDT's mutual dependence perspective has often been used in contexts where "firm autonomy (fear of dependency) is no longer the key goal of firms in acquiring external resources" (Shiu et al., 2023). Meta-organizations or other forms of decentralized multi-actor networks tend to focus on persuasion to commit others to exchange resources rather than using power (Shiu et al., 2023). Few actors alone are unlikely to possess all of the resources needed to identify modern slavery, demonstrating the need to acquire important resources held by others (Pfeffer and Salancik, 1978).

5.3. Practical implications

This study provides implications for a range of actor groups such as managers at firms, as well as for auditors, other service providers (i.e., organizations providing audit platforms), NGOs, and policymakers. For example, managers should seek applications of context-appropriate technology to identify modern slavery throughout their supply chains, noting that success will be contingent on obtaining resources held by a wide range of actors. This will require creating connections and transitions across the micro- and macro-levels of collaboration. Moreover, collaborations supporting (and supported by) technology could help reduce the 'psychological distance' of modern slavery. For example, psychological distance can be reduced with access to timely, high veracity information that "help bring events of persons perceptually 'closer' to a decision-maker" (Simpson et al., 2021). Worker voice technologies, for instance, can enable real-time data on labor conditions that help bring the experiences of workers that are typically far removed spatially and socially from managers, policymakers, or customers.

Beyond managers, there are many related implications for auditors and other service providers. While using monitoring technology is not about replacing human auditors, there are many ways it can impact how auditors perform their jobs. Auditor training, for example, must reflect the greater range of potential sources of evidence provided through technological applications (Castka and Searcy, 2023). Moreover, the boundary spanning activities across the conditions- and contracts-focused domains enabled by collaboration, technology, and other resources could provide a key differentiator in the services offered by different auditors (Cossio and McNeill, 2021).

There are also several related implications for other actors, including governments and NGOs. Governments, for example, should continue to enact legislation focused on modern slavery in supply chains, while being mindful that legislation alone will not identify it. Governments can take additional action, such as establishing trusted hotlines and supporting technological investments (such as through tax incentives) intended to identify modern slavery. Different NGOs will clearly have

different areas of focus and hold different resources, but examples could include focusing on enabling local access to communities, standardizing data collection protocols, providing trusted platforms for data exchange, and developing hotspot identification capabilities. Critically, all actors must recognize the challenges of identifying modern slavery working on their own, and that this goal is best achieved by explicitly recognizing mutual dependencies and actively working with others to build collaborations.

5.4. Limitations and future research

Building on RDT, this research illuminated the interdependencies between actors and resources through original empirical evidence gathered in a modern slavery context. However, our study also had limitations, which provides insight into several potential areas for future research.

Our study used experts from multiple settings and the results should be seen as an aggregated framework of the shared experiences of these actors. Our results thus describe the circumstances under which multi-actor collaborations typically leverage monitoring technologies to achieve their strategic goals. The conceptual framework is explicitly presented as a building block for future work. Additionally, we did not attempt to identify any 'weight' that should be allocated against each of the four key contingencies we identified. Further research on the relative importance of different contingencies in different contexts (e.g., in terms of sector/region/types of supply chains/types of technologies) would be valuable. Locational differences, for example, are an important dimension that shape resource dependencies in unique ways including, for example, rates of corruption, historical or political ties, and the level of trade dependence between countries (Jiang, H. et al., 2023). This means some variance in both the nature and effectiveness of collaboration across modern slavery contexts is likely given different country-specific dependencies. Research that explores such nuances will further our understanding of this important area.

Our study focused on identifying modern slavery. This, however, is only the start of efforts to address modern slavery in supply chains. Future research should focus on how modern slavery can be reduced and eliminated following its identification. The role of collaboration in facilitating the remediation of modern slavery is underexplored. While eliminating modern slavery in practice will likely require many key actors working together (McLaren et al., 2024), further underlining the applicability of RDT's mutual dependency perspective, it is also likely to require a strong understanding of power imbalance in the supply chain. Our study did not address the power imbalance perspective of RDT, but it could be an additional approach for considering actions to eliminate modern slavery. In particular, future studies can explore the circumstances under which actors are coerced into particular domain membership and whether such coercive mechanisms would change the dynamics of multi-actor collaboration, i.e., in general or specifically in terms of the choice of technologies adopted.

Finally, case studies on real-world applications of monitoring technologies in a modern slavery context are needed. This could include on-the-ground investigations of the benefits and challenges of using technology in different modern slavery settings. This would also provide further opportunity to study the contingencies in leveraging monitoring technologies through multi-actor collaboration, as well as an opportunity to study how collaborations evolve over time.

6. Conclusion

One of the reasons modern slavery persists is that it can be very difficult to identify in practice. Modern slavery is often embedded deep in supply chains far from public scrutiny. This creates a "psychological distance" between consumers, people experiencing modern slavery, and supply chain managers. Given its illegality in most jurisdictions, the prevalence of modern slavery in global supply chains is often actively

concealed. Even in cases where modern slavery is actively sought-out, it can be difficult to recognize. Technology is increasingly being used in supply chain management, and research on the use of technology in a modern slavery context is rapidly expanding (i.e., Kougkoulos et al., 2021; Christ and Helliär, 2021; Marques et al., 2024).

This study explored how multi-actor collaborations can leverage monitoring technologies to identify modern slavery in global supply chains. Modern slavery posture, collaboration scope, cross-boundary interactions, and contextual embeddedness individually or combined, can leverage (or inhibit) the likelihood of identifying modern slavery. These contingencies, which explicitly address our research question, represent four key contingencies in leveraging monitoring technologies through multi-actor collaboration, though their salience and impact can vary between supply chains. As demonstrated in an original conceptual framework, these themes are interconnected and can leverage monitoring technology through multi-actor collaborations to identify modern slavery through the contracts-focused and conditions-focused domains. While technology is not a panacea, we conclude that it can help identify modern slavery if it is supported by collaboration between employers, certification and auditing bodies, NGOs, and other vested intermediaries holding the financing, access, skills, and legitimacy to successfully apply it.

CRedit authorship contribution statement

Cory Searcy: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pavel Castka:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Grant Michelson:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaoli Zhao:** Writing – review & editing, Project administration, Investigation, Formal analysis, Data curation.

Declaration of competing interest

None.

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Data availability

The data that has been used is confidential.

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- Cory Searcy is a professor of industrial engineering at Toronto Metropolitan University (TMU). He currently serves as the Assistant Vice-President, International at TMU, as well as a corporate sustainability section editor at the *Journal of Business Ethics*.
- Pavel Castka is a professor of supply chain management and sustainability at Xi'an Jiaotong-Liverpool University (XJTLU), China. He currently serves as Associate Dean for Professional Engagement at International Business School Suzhou (IBSS) as well as committee member at TC176, International Organization for Standardization (ISO).
- Grant Michelson is a professor of management at Macquarie Business School, Sydney, Australia. His interests include corporate social responsibility and employee experiences of contemporary workplaces.
- Xiaoli Zhao is an assistant professor of strategic management at Xi'an Jiaotong-Liverpool University (XJTLU), China. Her interests include strategic aspects of networks providing plant-based dietary alternatives, wisdom leadership and international joint venture management.